Songyang Gao

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Research Interests

LLM-based Agents, Deep Research Agents, Auto-Research, AI for Science, Knowledge Graphs.

Education

The Hong Kong University of Science and Technology (Guangzhou) Sep 2025 – Jun 2027 (expected)
Master of Philosophy in Artificial Intelligence *Guangzhou, China*
- Advisor: Prof. Hui Xiong (IEEE, ACM, AAAI, AAAS Fellow)

Nanchang University (211, Double First-Class) Sep 2021 – Jun 2025
Bachelor of Engineering in Artificial Intelligence, Ji Luan Academy *Jiangxi, China*
- Baosteel Scholarship Special Prize, 2024 (**only 15 nationwide**) | Huawei Smart Base Scholarship (ratio: 0.3%)

Publications

Preprints / Under Review
- **S. Gao***, Y. Xia*, S. Liu, H. Xiong. "Graphs of Research: Citation Evolution Graphs as Supervision for Research Idea Generation." arXiv:2605.14790, 2026. (**First Author**) [Paper]
- Y. Xia*, **S. Gao***, H. Xiong. "SAB: A Streaming Agent Benchmark for Real-Time Data Pipelines." Under review at VLDB 2027. (**Co-first Author**)
- L. Gong*, T. Wang*, **S. Gao***, Y. Wu, Z. Hao, Y. Xu, L. Ding, H. Xiong. "Agentic Social Strategy Discovery: Learning What Matters from Branched Futures." Under review at AAAI 2027.

Published
- **S. Gao**, W. Xu, X. Wu, J. Liu, B. Wang, T. Wu, W. Wan, Q. Liu. "DMEDH: Diffusion Model-boosted Multiplane Extrapolation for Digital Holographic Reconstruction." **Optics Express**, 2024, 32(18):31920–31938. (CAS Tier 2, JCR Q2, **First Author**) [Paper]
- L. Zhang*, **S. Gao***, M. Tong, Y. Huang, Z. Zhang, W. Wan, Q. Liu. "HoloDiffusion: Sparse Digital Holographic Reconstruction via Diffusion Modeling." **Photonics**, 2024, 11(4):388. (JCR Q3, **Co-first Author**) [Paper]
- Z. Chen, B. Lin, **S. Gao**, W. Wan, Q. Liu. "Imaging through scattering media via generative diffusion model." **Applied Physics Letters**, 2024, 124: 051101. (JCR Q2) [Paper]
- L. Chen, Y. Zhou, **S. Gao**, M. Li, H. Tan, Z. Wan. "ARA-Net: An Attention-Aware Retinal Atrophy Segmentation Network Coping With Fundus Images." **Frontiers in Neuroscience**, 2023, 17: 1174937. (JCR Q2) [Paper]

Patents

- H. Xiong, Y. Xia, **S. Gao**. "Intelligent Encapsulation and Controlled Release System for Big Scientific Data Based on Multi-Stage Workflow and Stream Computing." (2026, Accepted)
- W. Wan, **S. Gao**, B. Wang, W. Xu, Q. Liu. "A Dual-Channel Prior-Constrained Multi-Distance Lensless Digital Holographic Reconstruction Method." ZL 2024107267996. (Student First Author, **Granted**)
- W. Wan, **S. Gao**, Z. Zhang, W. Xu, L. Zhang, Q. Liu. "A Sparse Digital Holographic Image Reconstruction Method." ZL 202410464769.2. (Student First Author, **Granted**)

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) Sep 2025 – Present
MPhil Researcher *Guangzhou, China*
Advisor: Prof. Hui Xiong (IEEE, ACM, AAAI, AAAS Fellow).
- **Deep Research System for Big Scientific Data** (From National Key Research and Development Program of China). Researching and designing Agentic Systems for Large-Scale Data Tasks, including building a Streaming Agent Benchmark for Real-Time Data Pipelines. As a co-first author, I completed the construction of the data execution pipeline and benchmarks, as well as the experimental testing. Outcomes include a paper under review at VLDB 2027, 1 filed patent, SciFlow and DeepScience.
- **Research Idea Generation via Graph Supervision (GoR).** Proposed Graphs of Research, a supervised fine-tuning method that uses citation evolution graphs as supervision for LLM-based research idea generation. GoR-SFT achieves SOTA against gpt-4o-driven baselines across head-to-head LLM-judge tournaments. [Paper]
- **Autonomous Research Agents (Redigg).** Built Redigg, an autonomous research-agent system for scientific discovery that combines a tool-calling loop, a modular skill registry, memory storage, and a React dashboard for streaming reasoning and tool traces. Implemented core research skills for paper search, PDF reading, literature synthesis, gap analysis, and report export, with HTTP/SSE and A2A interfaces for integration. [Code]

LIVE – Laboratory of Imaging and Vision Expression, Nanchang University Oct 2022 – Jun 2025
Research Assistant *Jiangxi, China*
Advisors: Prof. Qiegen Liu and Wenbo Wan

- Explored the role of diffusion models in computational imaging, aiming to improve the quality of image reconstruction by extracting prior information. The proposed method can improve the quality of digital holographic image reconstruction while saving the cost of optical devices.
- Mainly responsible for technical solution design, code implementation, simulations, real-world experiments, and manuscript writing. Developed diffusion-prior methods for digital holographic reconstruction, resulting in 3 journal papers, 2 granted patents, and a best-poster award.

Beijing Aishi Future Technology Co., Ltd. Mar 2025 – Aug 2025
AI Algorithm Research Intern *Beijing, China*

- Researched skeleton retargeting for human animation generation, enabling cross-identity motion transfer driven by pose sequences from diffusion-based video generation models.
- Conducted data processing, model training, and quantitative evaluation across multiple baselines including RealisDance, Animate-X, and MTVCrafter.

Institute for AI Industry Research (AIR), Tsinghua University Jan 2024 – Mar 2024
Research Intern (Winter Camp) *Beijing, China*
Supervisor: Yan Wang

- Based on the Stable diffusion model, accurate text-guided editing is performed. The generated diverse weather scenarios are used as datasets to train models for autonomous driving tasks.
- Mainly responsible for reproducing the open source project of the stable diffusion model and modifying it into a model suitable for weather scenarios. I improved my ability to use deep learning frameworks and gained a deeper understanding of editable diffusion models.

Professional Experience

Singtel PIE Program Dec 7–14, 2025
Group Leader *Singapore*

- Designed and proposed **SmartShield**, a multi-modal scam detection and prevention system via rule-LLM synergy, combining field research in Singapore with AI-driven solutions for real-world fraud prevention.

Tianjin Zhishan Information Technology Co., Ltd. Dec 2022 – Sep 2025
Technical Lead *Tianjin, China*

- Built a college student growth community with 700+ paying users. Developed RAG-based LLM applications, domain-specific intelligent agent Q&A systems (junyisocial.com), and brand website (zhishanzhe.com)

Honors & Awards

- China International College Student Innovation Competition, National Silver Award Oct 2024
- The Best Poster Award of the 6th Symposium on Computational Imaging Technology and Applications May 2024
- Outstanding Academic Correspondent in CAAI Transactions on Intelligence Technology (IF:7.985) May 2024
- The 14th "Challenge Cup" National Entrepreneurship Competition, National Bronze Award Nov 2024
- The 1st prize in the 5th Global Campus Artificial Intelligence Algorithm Elite Competition Dec 2023
- The 1st Prize in the 14th Blue Bridge Cup Special Competition Nov 2023
- The H Prize (2nd Prize) in the USA Mathematical Contest in Modeling Jul 2023
- The 3rd Prize in the 25th China Robot and Artificial Intelligence Competition May 2023
- The 1st Prize in the 13th MathorCup College Mathematical Modeling May 2023
- The 1st Prize in the Big Data Competition at the 3rd MathorCup College Mathematical Modeling Mar 2023

Selected Open-Source Projects

- **ds-spec-loop.** Open-sourced a reusable Spec-programming workflow that structures requirement clarification, architecture decisions, specification, task decomposition, verification, and convergence across code, tests, and documentation. [Code]

Skills

- **Programming & Data Systems:** Python, PyTorch, C/C++, SQL, Apache Flink, FastAPI, Next.js
- **AI/ML:** LLM-based Agents, LLM Fine-tuning, Diffusion Models, Computer Vision
- **Tools:** Docker, Git, Linux, vLLM, Slurm, Cloud Deployment